

HERS-VERIFIED MULTIFAMILYCENTRAL HOT WATER SYSTEM DISTRIBUTION



CALIFORNIA ENERGY COMMISSION

CEC-NRCV-PLB-21-H

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF VERIFICATION

Note: This table completed by **HERSECC** Registry.

Project Name:	Enforcement Agency:
Project Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information

01	Building Name	
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B. Design **HERS**-Verified Central Water Heating Systems Information (other than HPWH)

This table reports features of the water heating system other than **HPWH** system that were specified on the NRCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Water Heaters in System	Water Heater Storage Volume (gal)	Fuel Type	Rated Input Type	Rated Input Value	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value

B2. Design **HERS**-Verified CHPWH System Information

This table reports the water heating systems specified on the NRCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Modeled Equipment Make and Model	Number of Water Heaters/Compressors	Primary Tank Location	Primary Tank Volume	Primary Tank Total Insulation	Loop Tank Location	Loop Tank Volume	Loop Tank Total Insulation	Loop Pipe Insulation Thickness	Loop Tank Type	Simulated Equipment Make and Model

C. Installed **HERS**-Verified Central Water Heating Systems Information

This table reports the water heating system features other than **CHPWH** systems that were specified on the NRCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Water Heaters in System	Water Heater Storage Volume (gal)	Fuel Type	Rated Input Type	Rated Input Value	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value
13	Compliance Statement										

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C2. Installed HERS-Verified CHPWH System Information

This table reports the water heating systems specified on the NRCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10	11
Water Heating System ID or Name	Modeled Equipment Make and Model	Number of Water Heaters/ Compressors	Primary Tank Location	Primary Tank Volume	Primary Tank Insulation	Loop Tank Location	Loop Tank Volume	Loop Tank Insulation	Loop Pipe Insulation Thickness	Loop Tank Type

D. Design HERS Verified Central Water Heating Distribution Systems Information

This table reports the water heating distribution types specified on the NRCC compliance document for this project.

01	02	03	04	05	06
Water Heating System ID or Name	Central DHW System Distribution Type	Dwelling Unit DHW System Distribution Type	California Plumbing Code Appendix M	Master Mixing Valve	Insulation Verification

E. Installed HERS-Verified Central Water Heating Distribution Systems Information

This table reports the water heating distribution types specified on the NRCC compliance document for this project.

01	02	03	04	05	06
Water Heating System ID or Name	Central DHW System Distribution Type	Dwelling Unit DHW System Distribution Type	California Plumbing Code Appendix M	Master Mixing Valve	Insulation Verification

F. Installed HERS-Verified Water Heater Manufacturer Information

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number

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G. Mandatory Requirements for All Central Domestic Hot Water Systems

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	On systems that have a total capacity greater than 167,000 Btu/hr, outlets that require higher than service water temperatures as listed in the ASHRAE Handbook have separate remote heaters, heat exchangers, or boosters to supply the outlet with the higher temperature. (Section 110.3 (c)1)
02	Systems with circulating pumps or with electrical heat trace systems shall be capable of automatically turning off the system. (Section 110.3(c) 2).
03	Unfired storage tanks are insulated with: • External insulation of R-3.5, or • Internal insulation of R-16, or • The heat loss of the tank surface based on an 80°F water-air temperature difference shall be less than 6.5 Btuh/ft². (Section 110.3(c)34).
04	Recirculation loop shall meet the following requirements: • The recirculation pump is mounted on a vertical section of the return line, OR an automatic air release valve is installed on a riser at least 12 inches in length, on the inlet side of the recirculation pump, no more than 4 feet from the pump. (Section 110.3(c) 4A). • <u>A check valve or similar device shall be located between the recirculation pump and the water heating equipment to prevent water from flowing backwards through the recirculation loop.</u> A check valve is located between the recirculation pump and the water heater. (Section 110.3(c) 4B). • A hose bib is installed between the pump and the water heating equipment with an isolation valve between the hose bib and the water heating equipment. 110.3(c) 4C). • Isolation valves shall be installed on both sides of the pump, of which the item C valve can be one. 110.3(c)4D • The cold water piping and the recirculation loop piping shall not be connected to the hot water storage tank drain port. 110.3(c)4E • A check valve shall be installed on the cold water supply line between the hot water system and the next closest tee on the cold water supply line. 110.3(c)4F.
05	Instantaneous water heaters with an input greater than 6.8 kBTU/hr. (2kW) shall have isolation valves on both the cold water supply and the hot water line. (110.3 (c) 6).
06	<u>Domestic hot water piping insulation requirements (Section 150(J)):</u> • <u>All domestic hot water piping shall be insulated as specified in Section 609.12 of the California Plumbing Code. Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve.</u> • <u>Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated.</u> • <u>Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration.</u> • <u>Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members.</u> • <u>Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation.</u> • <u>Insulation is not required on the cold water line when it is used as the return</u> All sections of the recirculation loop, and the first 5 feet of all branches off the loop are insulated, to the thicknesses required by Table 120.3A. Other hot water piping shall meet the requirements of 150(j) and the installation requirements in, except for the following: (RA4.4.1): Piping in walls interior or exterior walls that is surrounded on all sides by at least 1 inch (2.5 cm) of insulation need not be insulated. • Piping installed in attics with a minimum of 4 inches (10 cm) of attic insulation on top • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. Metal piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating

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	material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. Insulation is not required on the cold water line when it is used as the return	
07	Domestic hot water piping insulation requirements: See the exceptions to Section 160.4(e) All piping for multifamily domestic hot water systems shall be insulated and meet the applicable requirements below: 1. General Requirements: - The first 8 feet of inlet cold water piping from the storage tanks, including piping between a storage tank and a heat trap shall be insulated. - Insulation on the piping and domestic hot water system appurtenances shall be continuous. - Pipe supports, hangers, and pipe clamps shall be attached on the outside of rigid pipe insulation to prevent thermal bridges. - All pipe insulation seams shall be sealed. - Insulation for pipe elbows shall be mitered, preformed, or site fabricated with PVC covers. - Insulation for tees shall be notched, preformed, or site fabricated with PVC covers. - Extended stem isolation valves shall be installed. - All plumbing appurtenances on hot water piping from a heating source to heating plant, at the heating plant, and distribution supply and return piping shall be insulated to meet the following requirements: - Where the outer diameter of the appurtenance is less than the outer diameter of the insulated pipe that it is attached to, the appurtenance shall be insulated flush with the insulation surrounding the pipe. - Where the outer diameter of the appurtenance is greater than the outer diameter of the insulated pipe that it is attached to, the appurtenance shall be insulated with a minimum thickness of 1 inch. - The insulation shall be removable and re-installable to ensure maintenance or replacement services can be completed. - Valves shall be fully functional without impediment from the insulation. 2. Insulation Thickness: All piping for multifamily domestic hot water systems shall meet the insulation thickness requirements specified in of Table 160.4-A. - For insulation conductivity in the range shown in Table 160.4-A for the applicable fluid temperature range, the insulation shall have the applicable minimum thickness or R-value shown in Table 160.4-A. - If the insulation conductivity falls outside the range provided in Table 160.4-A applicable fluid temperature range, the insulation shall meet a minimum R-value as indicated in Table 160.4-A. Or, it can have a thickness determined using Equation 160.4-A. - Insulation conductivity shall be determined in accordance with ASTM C335 at the mean temperature listed in Table 160.4-A, and shall be rounded to the nearest 1/100 Btu-inch per hour per square foot per °F. 3. Insulation Protection: Pipe Insulation shall be protected from damage due to sunlight, moisture, equipment maintenance and wind. Protection shall, at minimum, include the following: - Pipe and appurtenance insulation exposed to weather shall be protected by a cover suitable for outdoor service. The cover shall be water retardant and provide shielding from solar radiation that can cause degradation of the material. Appurtenance insulation covers shall be removable and able to be reinstalled. Adhesive tape shall not be used to provide this protection. - Pipe insulation covering chilled water piping and refrigerant suction piping located outside the conditioned space shall include, or be protected by, a Class I or Class II vapor retarder. All penetrations and joints shall be sealed. - Pipe insulation buried below grade must be installed in a waterproof and noncrushable casing or sleeve.	
087	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
098	Correction Notes:	

H. California Plumbing Code Appendix M

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	All distribution piping shall be sized according to the methodology specified in the California Plumbing Code Appendix M.
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H.I. ~~HERS-~~ Verified Multiple Dwelling Units Master Mixing Valves Installation Requirements ~~Verified Multiple Recirculation Loops for DHW Systems Serving Multiple Dwelling Units Requirements~~

For central systems with hot water piping serving multiple dwelling units master mixing valves (MMV) shall meet the following minimum specification, installation, and startup requirements specified in RA4.4.19.

~~All distribution systems listed on this form shall comply with these requirements.~~

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	All buildings with 8 or more dwelling units have a minimum of 2 recirculation loops.
02	Each loop roughly serves the same number of dwellings.
01	<u>Plumbing Plans:</u> The plumbing plans shall include the following MMV specification at a minimum: - Manufacturer's installation and commissioning instructions and plumbing drawings. - MMV conforms to the American Society of Sanitation Engineers (ASSE) 1017-2009 standard, Performance Requirements for Temperature Actuated Mixing Valves for Hot Water Distribution Systems. - Water mixing parameters and associated values: - Input parameters - Recirculation pump flow rate - Mixing valve outlet water temperature - Recirculation return water temperature - Mixing valve hot inlet water temperature - Calculated parameters - Percentage of water flow returning to cold side of MMV - Percentage of water flow returning to hot side of MMV - Manufacturer's operating parameter Maximum water mixing ratio
02	<u>Installation:</u> Installation of MMV shall meet manufacturer's instruction and the following requirements at a minimum: - The MMV shall be installed on the central heating plant hot water supply outlet header leading to the recirculation loop. - Check valves installed on the recirculation return line and cold-water line to inlet cold connection of MMV and on recirculation return piping leading back to storage tank or water heater. - Isolation valves installed on the inlet cold water, inlet recirculation return, inlet hot and outlet connections to MMV and on recirculation return piping connection to storage tank or water heater. - Balancing valve installed on the recirculation return piping to the water heater for MMVs that cannot 100% close the hot inlet port during operation. Thermometers installed on the outlet of the MMV and on the recirculation return line next the water pump.
03	<u>Startup:</u> - Startup testing of MMV during recirculation only operation. - Close all hot fixtures in the domestic water system. - Ensure that the water heater is operational and idling with storage tank plumbed to the mixing valve and meeting the hot inlet temperature specified in the plumbing plans. - Start the recirculation pump and set mixed outlet temperature or setpoint temperature on the MMV. Start the circulation pump at the specified water flow rate and adjust as needed to meet recirculation return temperature specified in the plumbing plans. - Let distribution system warm up and stabilize for 30 minutes and adjust mixing parameters as needed to realign with values in plumbing plans. - Let the recirculation pump operate for three hours without any water draws to ensure there is no temperature creep. - If during or after the three-hour period the MMV outlet and return temperature stays elevated by greater than 2°F and doesn't return back to the specified temperature, then make necessary adjustments to the MMV. If temperature creep persists with mechanical MMV, adjust the balancing valve as necessary on the recirculation return line leading back to the water heater to

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	<u>ensure average MMV outlet temperature meets the specified temperature.</u> <u>7. If adjustments are made to MMV or balancing valve in Step 6, then repeat Step 5.</u> b. Startup testing of MMV for a combination of recirculation and hot water draws. <u>1. Once the MMV is operational in a closed loop, make a water draw for 10 minutes using one of the following options:</u> <u>A. With a shower operating at full flow at every: three dwelling units in a building with 15 or fewer dwelling units, five dwelling units in a building with 16 to 30 dwelling units, eight dwelling units in a building with 31 to 60 dwelling units, ten dwelling.</u> <u>B. The hot water valve on a hose bib, mop sink, or other fixture on the branch line or location on the hot water distribution line is opened to a draw volume of 1 gpm for every: three dwelling units in a building with 15 or fewer dwelling units, five dwelling units in a building with 16 to 30 dwelling units, eight dwelling units in a building with 31 to 60 dwelling units, ten dwelling units in a building than 60 to 200 dwelling units, twenty dwelling units in a building with more than 200 dwelling units.</u> <u>2. Monitor recirculation return temperature on the thermometer during the 10-minute draw period and ensure design return water temperature is maintained at the specified temperature documented in the plumbing plans.</u> <u>If the recirculation return temperature falls more than 5°F below the specified temperature during the draw period, then adjust MMV setup to ensure compliance.</u>	
043	Verification Status:	☐ <u>Pass</u> - all applicable requirements are met; or ☐ <u>Fail</u> - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ <u>All N/A</u> - This entire table is not applicable
054	Correction Notes:	

J. Verified Pipe Insulation for Central System

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	<u>The heating plant and recirculation system piping insulation installation quality shall be field verified by a ECC-rater. The ECC-rater shall inspect the heating plant and horizontal supply header and return piping in accordance with mandatory requirements in Title 24 Part 6 section 160.4. The rater shall use a sampling approach that one in seven DHW recirculation pipe risers and associated branches be inspected to verify the pipe insulation meet with the following requirements:</u> <u>(a) All piping for multifamily domestic hot water systems shall be insulated to the thickness specified in Table 160.4-A, including the first 8 feet of inlet cold water piping to the heating plant. Insulation on the piping and appurtenances shall be continuous.</u> <u>(b) All appurtenances at the heating plant, from a heating source to storage tank(s), or in between storage tanks and storage water heaters, and recirculation supply and return loop shall meet the following:</u> <u>1. Insulation to be flush with pipe insulation or have minimum of one inch if appurtenance is bulkier.</u> <u>2. Removable and re-installable for maintenance or replacement.</u> <u>3. Pipe supports, hangers, and clamps shall be attached on the outside of rigid pipe insulation.</u> <u>(c) All pipe insulation seams shall be sealed along the length of the pipe and between adjacent sections of insulation material.</u> <u>(d) Insulation for pipe elbows shall be mitered, and insulation for tees shall be notched. Alternatively, tees and elbows may be pre-formed, or site fabricated with PVC covers.</u> <u>(e) Isolation valves shall be fully functional. Extended stem isolation valves shall be installed on hot water piping or where pipe insulation is required.</u>
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K. Determination of HERS-ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	
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Registration Number:

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DOCUMENTATION AUTHOR'S DECLARATION STATEMENT	
1. I certify that this Certificate of Verification documentation is accurate and complete.	
Name:	Signature:
Company:	Date:
Address:	CEA / HERSECC Certification Identification (if applicable):
City/State/Zip:	Phone:
RESPONSIBLE PERSON'S DECLARATION STATEMENT	
I certify the following under penalty of perjury, under the laws of the State of California:	
1. The information provided on this Certificate of Verification is true and correct. 2. I am the certified HERSECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater). 3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require HERSECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Nonresidential Appendices NA1, and NA2, NA7 and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency. 4. The information reported on applicable sections of the Certificate(s) of Installation (NRCI), signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (NRCC) approved by the enforcement agency. 5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement. 6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish this requirement.	
BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION	
Company Name (Installing Subcontractor or General Contractor or Builder/Owner):	
Responsible Builder/Installer Name:	CSLB License:
HERSECC PROVIDER DATA REGISTRY INFORMATION	
Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable)
HERSECC RATER INFORMATION	
HERSECC Rater Company Name:	
Responsible Rater's Name:	Responsible Rater's Signature:
Responsible Rater's Certification Number w/ this HERSECC	Date Signed:

NRCV-PLB-21-H User Instructions**A. General Information**

This table reports the building name - user input~~location as specified on the NRCC.~~

B. Design ~~HERS~~ Verified Central Water Heating Systems Information

This table reports features of the water heating system other than HPWH system - user input~~that were specified on the NRCC compliance document for this project. — For information only and requires no user input.~~

B2. Design ~~HERS~~ Verified CHPWH System Information

This table reports the water heating systems - user input~~specified on the NRCC compliance document for this project.~~

C. Installed HERS Verified Central Water Heating Systems Information

This table reports the water heating system information that is being installed. Require one line for each central system.

1. Water Heating System ID or Name – Reference information from Table B~~NRCC~~.
2. Water Heating System Type – Reference information from Table B~~NRCC~~. The different kinds of water heating system type are DHW or Combined Hydronic.
3. Water Heater Type – Information from Table B~~NRCC~~. The different kinds of water heaters are Large/Commercial Storage, Small/Consumer Storage, Residential-Duty Commercial Storage, Heat Pump, Boiler, Large/Commercial Instantaneous, Small/Consumer Instantaneous, Residential-Duty Commercial Instantaneous or Indirect.
4. # of Water Heaters in system – Reference information from Table B~~NRCC~~.
5. Water Heater Storage Volume (gal) – User input. Value may be N/A if water heater type is instantaneous with zero storage.
6. Fuel Type – Reference information from Table B~~NRCC~~. The different kinds of fuel types are natural gas, propane, oil, or electricity.
7. Rated Input Type – Reference information from Table B~~NRCC~~. For natural gas, propane and oil fuel type the input type is Btu/Hr. For electric the input type is kW.
8. Rated Input Value – User input. Numerical value of the rated input. Must be equal to or less than value indicated on the Table B~~NRCC~~.
9. Heating Efficiency Type – Reference information from NRCC. Different efficiency types are Energy Factor, AFUE, UEF and Thermal Efficiency.
10. Heating Efficiency Value – User input. Numerical value of the Heating Efficiency. Must be equal to or higher efficiency than value indicated on the Table B~~NRCC~~.
11. Standby Loss – User input. Must be equal to or less than value indicated on the Table B~~NRCC~~. Value may be N/A if NRCC value is N/A.
12. Exterior Insul. R-Value – User input. Must be equal to or higher than value indicated on the NRCC. Value may be N/A if NRCC value is N/A.

C2. Installed **HERS**-Verified CHPWH System Information

This table reports the water heating system information that is being installed. Require one line for each installed water heater. Require one line for each installed water heater.

1. Water Heating System ID or Name – Reference information from Table B2.
2. Modeled Equipment Make and Model – User input must be equal to the value indicated on Table B2 as default and allow user to override with an equivalent system based on the simulated equipment in Table B2.
3. Number of Water Heaters/ Compressors – User input, must be equal to the value indicated on table B2.
4. Primary Tank Location – Reference information from Table B2.
5. Primary Tank Volume – User input, must be equal to or higher than the value indicated on table B2.
6. Primary Tank Insulation – User input, must be equal to or higher than value indicated on table B2.
7. Loop Tank Location – Reference information from Table B2.
8. Loop Tank Volume – User input, must be equal to or higher than the value indicated on table B2.
9. Loop Tank Insulation – User input, must be equal to or higher than value indicated on table B2.
10. Loop Pipe Insulation Thickness – User input, must be equal to or higher than the value indicated on table B2.
11. Loop Tank – Reference information from Table B2.

D. Design **HERS**-Verified Central Water Heating Distribution Systems Information

1. Water heating distribution types specified on the Table B or Table B2 ~~NRCC compliance document for this project.~~
2. User input
3. User input

E. Installed **HERS**-Verified Central Water Heating Distribution Systems Information

1. ~~01~~ Water Heating System ID or Name = Reference information from Table D.
2. ~~02~~ Central DHW System Distribution Type = Reference information from ~~NRCC~~ Table D.
3. Dwelling Unit DHW System Distribution Type = Reference information from Table D

F. Installed **HERS**-Verified Water Heater Manufacturer Information

This table reports the manufacturer information of the installed water heater(s). Require one line for each installed water heater.

1. Water Heating System ID or Name – Reference information from ~~NRCC~~ Table C.
2. Manufacturer – User input. Enter the name of the water heater manufacturer.
3. Model Number – User input. Enter the model number of the water heater.

G. Mandatory Requirements for All Central Domestic Hot Water Recirculation Systems

This table lists the requirements for all central recirculation systems. Installer must ensure all the requirements in this table are met.

H. California Plumbing Code Appendix M

This table lists the requirements for California Plumbing Code Appendix M. ECC rater must ensure all the requirements in this table are met.

~~H.I. HERS-Verified~~ Multiple Dwelling Units Master Mixing Valves Installation Requirements
~~Multiple Recirculation Loops for DHW Systems Serving Multiple Dwelling Units Requirements~~

Dwelling Units Master Mixing Valves. ECC rater must ensure all the requirements in this table are met. ~~This table applies to all systems identified on this compliance document. This measure requires on site HERS verification that at least two central recirculation loops are included in the system design. This credit is available to buildings with 8 or more units. The recirculation loops must be relatively equal in length and supply approximately the same number of dwelling units.~~

J. Verified Pipe Insulation for Central System

This table lists the requirements for Verified Pipe Insulation for Central System. ECC rater must ensure all the requirements in this table are met.

~~H.K.~~ Determination of HERS Verification Compliance

1. This field is filled out automatically. Compliance requires that all individual criteria pass.

A. General Information

01	Dwelling Unit Name	<<user input>>
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B. Design ~~HERS~~ Verified Central Water Heating Systems Information (other than CHPWH)

<<Static Text - see the top>><<If NRCC only has CHPWH, display the "section does not apply" message>>

<<require one row of data for each water heater identified on the NRCC-PRF>>

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Water Heaters in System	Water Heater Storage Volume (gal)	Fuel Type	Rated Input Type	Rated Input Value	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value
<< User input >>	<< Allow user to select from list (DHW, Hydronic, or Combined Hydronic) >>	<< Allow user to select from list: (Boiler, Indirect, Consumer Instantaneous, Commercial Instantaneous, Consumer Storage, Commercial Storage, Residential I-Duty Commercial Storage, or Residential I-Duty Commercial Instantaneous) >>	<< User input, value must be greater than or equal to 1>>	<< User input, if B03 = Consumer Instantaneous, Commercial Instantaneous, or Residential I-Duty Commercial Instantaneous then allow N/A; >>	<< User select from list (Btu/h, kW, N/A)>>	<< User select from list (Natural Gas or Propane, then value = Btu/Hr; Else if B06 = Electric Resistance, then value = kW >>	<<User input if B03 = Heat Pump, then result = NA; >>	<< User select from list (AFUE, Thermal Efficiency, Uniform Energy Factor)>>	<< User input, but allow N/A; >>	<< User input, but allow N/A; >>	<< User input, but allow N/A; >>

B2. Design ~~HERS~~ Verified CHPWH System Information

<<Static Text - see the top>><<If NRCC has CHPWH, then require table, else display the "section does not apply" message>>

<<require one row of data for each water heater identified on the NRCC>>

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Modeled Equipment Make and Model	Number of Water Heaters/ Compressors	Primary Tank Location	Primary Tank Volume	Primary Tank Total Insulation	Loop Tank Location	Loop Tank Volume	Loop Tank Total Insulation	Loop Pipe Insulation Thickness	Loop Tank Type	Simulated Equipment Make and Model
<< User input >>	<< User select from list (see NEEA HPWH Official List –	<< User input, value must be greater than or equal to	<< User select from list (Attic, Conditioned, Crawl Space,	<< User input >>	<< Default to R-16, but allow user to override and enter a higher value >>	<< User select from list (Attic, Conditioned, Crawl Space,	<< User input >>	<< User input >>	<< User input >>	<< User select from list (Commercial HPWH, Electric Resistance	<< hide column from user, do not require field>>

	CHPWH tab); >>	1>>	Garage, TankZone , Unconditi oned)>>			Garage, TankZone, Unconditi oned)>>				, Large Compress or, NEEA)>>	

C. Installed HERS-Verified Central Water Heating Systems Information

<<Static Text - see the top>><<If Table B does apply, then require table, else display the "section does not apply" message>>

<<require one row of data for each water heater identified in Table B>>

01	02	03	04	05	06	07	08	09	10	11	12
Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Water Heaters in System	Water Heater Storage Volume (gal)	Fuel Type	Rated Input Type	Rated Input Value	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value
<< Reference values from B01 >>	<< Reference values from B02 >>	<< Reference values from B03 >>	<< Reference values from B04 >>	<< Reference values from B05 >>	<< Reference values from B06 >>	<< Reference values from B07 >>	<<User input value which must pass the following range tests: If C06 = Heat Pump, then C08 = NA; If C06 = Natural Gas or Propane, then If C03 = Commercial Storage, then value must be > 75,000 Btu/hr; If C03 = Consumer Storage, then value must be ≤ 75,000 Btu/hr; If C03 = Commercial Instant, then	<< Reference values from B09; If prescriptive, then user select from: AFUE, Thermal Efficiency, Energy Factor>>	<< User Input must ≥ B10, value may only be NA if B10 = NA >>	<< User Input must ≤ B11, value may only be NA if B11=NA> >	<< User Input must ≥ B12, Value may only be NA if B12 = NA>>

							value must be > 200,000 Btu/hr; If C03 = Consumer Instant, then value must be ≤ 200,000 Btu/hr; Else if C03 = Residential I-Duty Commercial Storage, then value must be ≤ 105,000 Btu/hr; Else if C06 = Electric Resistance, then If C03 = Commercial Storage or Commercial Instant, then value must be > 12 kW; If C03 = r Consumer Storage or Consumer Instant, then value must be ≤ 12 kW; Else if C03 = Residential I-Duty Commercial Instantan eous, then value must be ≤ 58.6 kW; End If				
--	--	--	--	--	--	--	---	--	--	--	--

							If the value passes range test, it is stored in WaterHeaterElectricFiredRateInput, if C06 = Electric Resistance. Otherwise the value is stored in WaterHeaterGasFiredRateInput; Elseif C03 = Boiler or Indirect, no limit on input value >>>>				
13	Compliance Statement			<<calculated field: If C08 ≤ B08, and C10 ≥ B10, and C11 ≤ B11, and C12 ≥ B12, then display result = System complies; else display result = system does not comply>>							

C2. Installed ~~HERS~~ Verified CHPWH System Information

<<Static Text - see the top>><<If Table B2 does apply, then require table, else display the "section does not apply" message>><<require one row of data for each water heater identified in Table B2>>

01	02	03	04	05	06	07	08	09	10	11
Water Heating System ID or Name	Modeled Equipment Make and Model	Number of Water Heaters/ Compressors	Primary Tank Location	Primary Tank Volume	Primary Tank Insulation	Loop Tank Location	Loop Tank Volume	Loop Tank Insulation	Loop Pipe Insulation Thickness	Loop Tank Type
<<reference values from B2-01>>	<< If performance, user input is equal to B2-02 as default, and allow user to override with an equivalent system based on the simulated equipment in B2-12 >>	<< User Input must equal reference values from B2-03 >>	<< Reference values from B2-04>>	<<User input, must be ≥ value from B2-05>>	<< User input, must be ≥ value from B2-06>>	<< Reference values from B2-07>>	<<User input, must be ≥ value from B2-08>>	<< User input, must be ≥ value from B2-09>>	<< User input, must be ≥ value from B2-10>>	<<Reference value from B2-11>>

12	Compliance Statement:	<<calculated field: If B2-08 ≤ C2-08, and B2-10 ≥ C2-10, and B2-11 ≤ C2-11, and B2-12 ≥ C2-12, then display result = System complies; else display result = system does not comply>>
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D. Design ~~HERS~~ Verified Central Water Heating Distribution Systems Information

<<Static Text - see the top>><<require one row of data for each water heating system identified on the LMCC-PRF>>>

01	02	03	04	05	06
Water Heating System ID or Name	Central DHW System Distribution Type	Dwelling Unit DHW System Distribution Type	<u>California Plumbing Code Appendix M</u>	<u>Master Mixing Valve</u>	<u>Verified Insulation</u>
<<reference values from LMCC (see rule in header)>>	<<Reference value from LMCC-PRF-01. Allowed values are: * Multi-family: Recirculation * Multifamily: Non-recirculation>>	<<Reference value from LMCC-PRF-01. Allowed values are *Standard Distribution System * HERS -Verified <u>Central</u> Pipe Insulation Else if Prescriptive, value = *Standard Distribution System>>	<<If Performance, reference value from <u>LMCC-PRF-01</u> . Allowed values are *Yes *No Else if Prescriptive, value = *Yes>>	<<If Performance, reference value from <u>LMCC-PRF-01</u> . Allowed values are *Digital *Mechanical *None Else if Prescriptive, value = *Digital or Mechanical>>	<<If Performance, reference value from <u>LMCC-PRF-01</u> . Allowed values are *Yes *No Else if Prescriptive, value = *Yes>>

E. Installed ~~HERS~~ Verified Central Water Heating Distribution Systems Information

<<Static Text - see the top>><<require one row of data for each water heating system identified on the LMCC-PRF>>

01	02	03	04	05	06
Water Heating System ID or Name	Central DHW System Distribution Type	Dwelling Unit DHW System Distribution Type	<u>California Plumbing Code Appendix M</u>	<u>Master Mixing Valve</u>	<u>Verified Insulation</u>
<<reference values from D01>>	<< Reference value from D02>>	<< Reference value from D03>>	<< Reference value from D04>>	<< Reference value from D05>>	<< Reference value from D06>>

F. Installed ~~HERS~~ Verified Water Heater Manufacturer Information

<< require one row of data for each water heater identified in Section C, elseif no Section C, then display the "section does not apply" message >>

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number
<< Reference value from C01>>	<<User input>>	<<User input>>

G. Mandatory Requirements for All Central Domestic Hot Water Systems

<<Static Text - see the top>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

H. California Plumbing Code Appendix M

<< Display if any water heater system identified in D04=Yes, elseif D04=no in all rows, then display the "section does not apply" message.>>

<<Static Text - see the top>>

H.I. HERS-Verified Multiple Dwelling Units Master Mixing Valves Installation-Recirculation Loops for DHW Systems Serving Multiple Dwelling Units Requirements

<<Static Text - see the top>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

J. Verified Pipe Insulation for Central System

<< Display if any water heater system identified in D06 = yes, elseif D04=no in all rows , then display the "section does not apply" message.>>

<<Static Text - see the top>>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met. Requirements

K. Determination of HERS-Verification Compliance

<<Static Text - see the top>>

- | | |
|----|---|
| 01 | << if C13, C2-12 and E12 result = System Complies, and results for all applicable sections I and J ≠ fail, then display: Complies: All specified verification protocol requirements on this document are met; else display: Does not comply: One or more specified verification protocol requirements on this document are not met.>> |
|----|---|